

H2GUARD · REQUIREMENTS

H2Guard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Status from HGD-CAL-001 v0.4 after the decisions were carried into the model and BOM; R1 placement met on the drop rod, R5 with the pressure switch, R13 not met on estimate (5.0 h), R14 USD 411
0.6	2026-10-02	Amish Chadha	R1 placement and R5 airflow proving from the decisions of 2026-10-02 (HGD-DEC-001); set point source rule
0.5	2026-10-01	Amish Chadha	Design for construction (HGD-DDR-003); R13 and R14 status from HGD-CAL-001 v0.3; R14 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R1 placement rule, R3 timed escalation, R13 scope and R14 budget restated; status from HGD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from HGD-CAL-001; set points and inventory rule adopted for TRL 3 per HGD-DDR-001 (pending Amish's review); R6 names the series

			relay and hardware latch; R12 names the bump test cup and port
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

H2Guard requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after sessions with the first users. At TRL 3 each requirement is checked by calculation in HGD-CAL-001 v0.4 against the design in HGD-PRC-001 v0.7, the constructable model in `cad/src/model.py` (HGD-DDR-003, with the decisions of 2026-10-02 carried in) and the build plan HGD-BLD-001. One requirement is **not met on its estimate**: R13 (installation, 5.0 h against 4 h, after the drop rod, pressure switch and their cables were added). Two are at risk: R4 (assumed sensor t90) and R7 (assumed fan curve). R14 is reported against the value-engineering target: the constructable design is estimated at USD 411, USD 146 over the USD 265 target. R1 cannot be verified without sensor data. On 2026-09-25 Amish accepted the review recommendations (HGD-DDR-002): R1 gains the head placement rule, R3 gains the timed escalation, R13 excludes the fan and grille wall openings as builder's work, and R14 follows the USD 265 figure, which is a value-engineering target, not a limit (Amish, 2026-10-01). The set points (R2, R3) and the inventory rule (R9) are decided by Amish (HGD-DDR-001, D2 and D5).

Table 1. Requirements. LFL is the lower flammability limit of hydrogen in air, taken as 4.0 % by volume, so 10 % LFL is 0.4 % vol and 25 % LFL is 1.0 % vol. Tags in brackets refer to lines of HGD-CAL-001.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (HGD-CAL-001)
R1	Measure hydrogen at the high point of the room	0 to 100 % LFL (0 to 4 % vol), resolution 1 % LFL or better; head within 0.3 m of the ceiling and directly above each likely leak point, with a second head where leak points are far apart; the horizontal offset limit is set at TRL 4 from the measured plume width	Sensor datasheet review; installation rule in the precis; offset limit by test at TRL 4	Not verifiable at TRL 3: resolution needs sensor data; height met (ports 175 mm below the ceiling [L1]); plume radius about 141 mm at the ports [E7]. Placement met: the head hangs from the ceiling on a drop rod with its ports 0 mm from the apparatus centre, inside the plume (decided 2026-10-02; the former wall position, 260 mm off, did not meet the rule)
R2	Warn early	At 10 % LFL (0.4 % vol): fan to boost, amber light, intermittent sounder; self-clearing	Set point and logic review	Met on paper: design leak plume at the head about 15 % LFL [E2]
R3	Trip and shut off the supply	At 25 % LFL (1.0 % vol): valve closed, fan on boost, continuous sounder and beacon; latched until reset with a key and the reading is below 10 % LFL. A warning held for 5 min also closes the valve and latches in the same way (firmware rule)	Set point and logic review	Met by design: on the plume axis the design leak trips (about 30 % LFL [E2]); off the axis the escalation closes the valve at 5.6 min with 28 L released [F7]
R4	Respond fast enough	Sensor t ₉₀ of 30 s or less; alarm within 60 s of a step to 1.1 % vol (twice t ₉₀ , following the ISO 26142 test approach); valve closed within 2 s of the trip	Datasheets, then a bench test at TRL 4	At risk: 35.7 s leak to valve closed with an assumed 30 s t ₉₀ ; 0.17 s trip to valve closed [F3, F4]
R5	Fail safe	The supply is closed within 2 s of any of: loss of 24 V power, sensor open or short circuit, heater failure, controller watchdog timeout, fan stopped (no tachometer pulses for 10 s), airflow not proven by the differential pressure switch at the fan (added 2026-10-02)	Failure modes and effects analysis (FMEA)	Met on paper: worst listed fault 1.12 s, airflow loss included [F6]; the pressure switch on a fan inlet tap detects a blocked duct or stuck shutter [G6]. Three dangerous undetected faults remain (HGD-CAL-001 Table 4): two covered by the bump test, and the leaking valve seat covered by the valve's hydrogen and seat leak ratings and the TRL 4 proof test
R6	Trip without firmware	A hardware comparator on the catalytic sensor bridge, with its own latch, opens a relay in series with the microcontroller's valve switch at the trip level, even if the microcontroller has stopped	Circuit review	Met by design (block diagram level)
R7	Ventilate the room	Continuous exhaust at high level of 5 air changes per hour or more (150 m ³ /h for the 30 m ³ reference room); boost of 10 air changes per hour or	Fan curve against duct and grille losses	At risk: 347 m ³ /h boost with the re-specified fan on an assumed curve; the TRL 2 fan gave 237 m ³ /h [G3]

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (HGD-CAL-001)
		more; make-up air at low level on the far side		
R8	Keep the design leak below the trip level	With the design leak of 5 L/min and continuous ventilation, the well-mixed room stays below 10 % LFL	First-order dilution calculation	Met on paper: 5.0 % LFL well mixed and in the upper layer [D1, D3]
R9	Limit what can leak	Hydrogen inventory that could be released into the room is 1 % of room volume or less at atmospheric pressure (300 L for 30 m ³), unless a flow restrictor limits release to the design leak	Installation rule and host project review	Met by design as an installation rule; H2Bench's tank is 2.6 % of the limit; a 0.13 mm orifice at 10 bar gauge limits a failure to 5 L/min [B3, B4]
R10	Alert people	Sounder 85 dB(A) or more at 1 m and a red beacon visible from the room entrance	Component ratings	Met on paper: 90 dB at 1 m, 76 dB at 5 m [K2]
R11	Keep mains out of the self-built parts	All self-built wiring 24 V DC; mains only inside a certified power supply; power draw 60 W or less	Design review	Met on paper: 48.7 W peak in the warning state, 49.2 W with the oxygen sensor option [H2, H3]
R12	Be checkable in use	Bump test with 1 % vol hydrogen span gas through the test port and head cup in 2 min or less; self-test button runs the alarm, fan boost and valve close; event log of readings, warnings, trips and resets kept for 90 days or more	Design review	Met on paper: bump reading settles in about 67 s [J1]; log 2.07 MB in 4 MB of flash [J3]; cup delivery factor unknown, so the pass level is set at TRL 4
R13	Install simply	Mounted and wired by one person with hand tools in 4 h or less, excluding the gas fitting, which a competent person makes, and the wall openings for the fan and make-up grille, which are builder's work	Task time estimate; walk-through at TRL 4	Not met on estimate: 5.0 h, 60 min over [M1] (6.9 h if the openings were included); was at risk (4.2 h) at v0.6
R14	Cost against the value-engineering target	Value-engineering target USD 265 (budget_usd , a hypothetical control target)	Priced BOM	Over the value-engineering target by USD 146: estimated cost of the constructable design USD 411 [N1] on indicative prices; options (oxygen sensor, alternative fan) priced separately [N3]
R15	Avoid adding ignition sources in the ceiling layer	Fan never switched on in a flammable mixture (runs continuously); controller, relays and power supply mounted at least 1 m below the ceiling; sensor ports behind sintered flame arrestors	Design review	Met by design: controller top 1,025 mm below the ceiling [L2]; not certified for hazardous areas, and the catalytic element runs hot

■ Assumptions

- Reference room: 4 x 3 x 2.5 m (30 m³), one door, general ventilation unknown and taken as zero.
- Design leak: 5 L/min of hydrogen, representing a failed fitting downstream of a regulator with a flow restrictor. A small electrolyzer of about 100 W makes only about 0.39 L/min (HGD-CAL-001, B1), so this covers it with margin. A failed regulator on an unrestricted cylinder can release far more, and H2Guard does not protect against that (R9).
- Hydrogen rises from the leak as a buoyant plume and collects under the ceiling. HGD-CAL-001 uses plume theory for the head and a well-mixed or displacement model for the room.
- Set points follow common practice for hydrogen detection (warning at 10 % LFL, shutdown at 25 % LFL); ISO 26142 describes alarm levels at 25 % or 50 % LFL. The set points are decided by Amish, 2026-09-25 (HGD-DDR-001, D2). On 2026-10-02 Amish decided that each site takes its set points and inventory rule from the stricter of the national code and ISO 26142, and from the host institution's safety office where its rules are stricter; 10 % and 25 % LFL stay the defaults.
- Value-engineering target: the USD 265 in `project.yaml` (raised from USD 180 by Amish's acceptance of the recommendation, HGD-DDR-002) is a hypothetical control target, not a limit. It covers the H2Guard kit including the fan and make-up grille, but not the gas system, span gas, wall openings or installation labor.